

Data Investigation Proposal

Unit 8: Sports Analytics Lab • Lesson 8-1 • Flagship • EduWonderLab

UNIT 8 • LESSON 8-1 ★ FLAGSHIP • TEACHER NOTES

Standard 6.SP.1

FORMAT Mini-survey design project	TIME 30–35 min
GROUPING Teams of 3–4	TOPIC Statistical Questions and Data (Flagship)
STANDARD 6.SP.1	PREP LEVEL Medium — print proposal sheet and class tally

Learning goal *I can design a statistical survey question, predict variability, and collect a small data set.*

Teacher setup

- Print one proposal sheet per team. Teams write a statistical survey question.
- They predict how much the answers will vary, then collect a small data set.
- Use the checklist to improve a weak question.

Materials

Survey proposal sheet, Class data tally, Question-improvement checklist, Pencil.

Differentiation & language support

- Checklist: Does it vary? Is it clear? Can you collect it? (SPED)
- Sentence stem: “My question is statistical because ____.”
- ESOL: give two example questions to model the task.

Key vocabulary (English / Español):

Term	Español	What it means
survey	encuesta	a way to collect data from people
variability	variabilidad	how much the answers differ

Quick teacher check: *Check that each team’s question produces varying answers.*

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Name:

Date:

Class:

Your goal: *I can design a statistical survey question, predict variability, and collect a small data set.*

What you need

Survey proposal sheet, Class data tally.

How to play

1. Write a statistical question about your class.
2. Predict whether the answers will vary a lot or a little.
3. Collect at least five responses and record them.
4. Explain why your question is statistical.

Proposal Steps

Design a statistical question, predict how the answers will vary, then collect a small data set.

1. Write a statistical question for your class.	2. Predict: will the answers vary a lot or a little?
3. Collect 5 responses. Record them.	4. Why is your question statistical?
5. Write a non-statistical version of your question.	6. Improve a weak question: "Do you like sports?"

Proposal (question, prediction, data, reason):

Explain your thinking

Explain how you know your data will help answer a statistical question.

Sentence frame: *My question is statistical because ____.*

★ **Challenge:** *Trade questions with another team and predict the variability of their data.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Write a statistical question for your class. Answer: e.g., “How many hours of sleep do students get?”	2. Predict: will the answers vary a lot or a little? Answer: A lot — that is what makes it statistical
3. Collect 5 responses. Record them. Answer: Sample: 6, 7, 8, 8, 9 hours (answers will differ)	4. Why is your question statistical? Answer: Because different students give different answers
5. Write a non-statistical version of your question. Answer: e.g., “How many hours did I sleep last night?” (one answer)	6. Improve a weak question: “Do you like sports?” Answer: e.g., “How many hours of sports do students play each week?”

Teaching notes & look-fors

- A good statistical question is clear, varies, and is collectable.
- Card 6: a yes/no question becomes statistical when it asks “how many/how much.”